$$D(M) = M \tag{2}$$

Consistent with prior work [56], we compute each model's perplexity on the original Wikitext2 test split to assess model performance. Perplexity reflects a model's ability to complete the test set text sequences (lower is better) and indicates the degree of alignment between the model's probability distribution p_θ and the true distribution p_{data}.

In an AI ecosystem composed of M different models, we calculate the combined mean perplexity μ for a given iteration step t as:

$$\mu_t = \frac{1}{M} \sum_{m=1}^{M} \mu_{m,t} \ , \tag{3}$$

whereas $\mu_{m,t}$ corresponds to the mean perplexity across the text sequences of the original Wikitext2 test split.

To compare ecosystems across iterations, we calculate an aggregated performance. Let μ_t an ecosystem's mean perplexity for a given iteration step t, then, the aggregated mean perplexity μ_T across T iterations equals:

$$\mu_T = \frac{1}{T} \sum_{t=1}^{T} \mu_t \tag{4}$$

Finally, for a fixed number of iterations T and a set of ecosystems with varying diversity levels, we define the optimal diversity level as the one that achieves the highest aggregated performance:

$$D_{\text{opt}}(T) = \arg\min_{D} \mu_T(D). \tag{5}$$

3.2 Generation and training details

We use OPT [72] and GPT2 [51], two open-accessible, small, and well-studied language model families used in prior work on collapse [e.g., 31, 56]. For the primary approach, we use the smallest versions, OPT-125m and GPT2, with 125M and 124M parameters, respectively.

Analogous to prior work [31, 56], we use Wikitext2 [45] as the original dataset. To reduce computational effort during training and generation, the primary approach uses a randomly sampled subset of two-fifths of the original size per ecosystem ($N = 0.8$M tokens), while keeping the original Wikitext2 validation and test splits for evaluation. Using a block size of 128 tokens, the sizes of the training–generation, validation, and test splits correspond to 7,500, 1,900, and 2,200 data points, respectively.

Our approach for generating the artificial dataset follows a method introduced in prior work [56]. For each available token sequence of length 128 in the generation dataset, we prompt the model to generate a continuation sequence of equal length, using a five-way beam search. Because we iterate through the entire generation dataset, the model would produce the original generation dataset if it had zero error. At each iteration, each model is trained for 5 epochs with a learning rate of $1e^{-5}$ on a training sample of size $n = N/M$, using a block size of 128 tokens. The best-performing model on the validation set is preserved for subsequent iterations.

To study the effect of increasing parameter size, we use OPT-350M with 350M parameters and the full Wikitext2 dataset (2.1M tokens); see Section 3.3. The analogous experiments with GPT2-Medium (355M parameters) are reported in Appendix F. All experiments are carried out using the HuggingFace Transformers library [67] and the Python Lightning framework [19].

3.3 Variations of primary setting

To understand the effect of diversity in more realistic settings, we examine how the location of the optimum shifts under two types of experimental variations: V1 examines the effect of increasing model and data sizes, and V2 examines the effect of data quality improvements through real-data incorporation and temperature sampling. We conduct the experimental variations using the OPT model family and report the scaling experiments with GPT2 in Appendix F.

V1: System scale –– model size and data size. In reality, both model and data sizes are far larger. OPT-125m contains 125M parameters, whereas current large language models (e.g., OpenAI's ChatGPT, Google's Gemini, Anthropic's Claude) likely contain in the order of trillions of parameters. A similar discrepancy applies to dataset size: while Wikitext2 contains roughly 2M tokens, the datasets used for prevailing models are likely 10^6 times larger. Established scaling laws suggest a power-law relationship between performance and parameter size [33, 37]. However, recent work indicates that these relationships may break down under self-training conditions and that larger models and datasets may even amplify collapse [16, 17]. To examine ecosystem diversity through the lens of scaling laws we introduce three variations of the primary setting: (1) a larger model (OPT-350m, with 350M parameters), (2) the full Wikitext2 training set (2.1M tokens), and (3) the combination of both larger model and dataset sizes. We conduct these experiments for M =1, 2, 4, 8, and 16 models.

V2: Training data quality –– real-data incorporation and sampling temperature. In realistic self-training loops, artificial data does not replace but rather accumulates alongside real (non-model-generated) data, improving the quality of the training set and mitigating collapse [28, 56, 70]. Similarly, prevailing language models are not deterministic but incorporate randomness that diversifies outputs and can mitigate model collapse [55, 68, 74]. A common mechanism is temperature sampling, which increases output diversity by flattening token-sampling probabilities [2, 12]. To test these factors, we introduce an experimental variation that replaces 10% of the model-generated data with randomly sampled original Wikitext2 training data at each iteration (following Shumailov et al. [56]), as well as three additional variations using sampling temperatures of $\tau = 0.5$, 1.0, and 2.0. As a reminder, the primary experiment uses deterministic generation with $\tau = 0$. When temperature sampling is applied, all remaining sampling parameters follow the Transformers default settings, corresponding to `top_k` = 50 and `top_p` = 1.0.

4 Results

We conducted a series of experiments to examine how diversity across models affects performance when models are trained on their collective outputs. Under a fixed data size and compute budget, we find that the optimal degree of diversity increases monotonically with the number of self-training iterations:

$$T_1 < T_2 \implies D_{opt}(T_1) \leq D_{opt}(T_2), \tag{6}$$

where the optimal level is defined as the ecosystem achieving the highest aggregated performance (i.e., lowest aggregated mean perplexity). In other words, as the number of self-training iterations increases, maximizing overall performance requires segmenting the data across more models. Correspondingly, the benefits of diversity relative to the most homogeneous setting grow with the number of iterations.

In the primary approach, the single-model setting ($D = 1$) performs best only in the short term (considering fewer than four iterations for OPT-125M and fewer than three iterations for GPT2, see Figure 3) Figure 4 shows the corresponding mean perplexity trajectories for ecosystems consisting of 1, 2, 4, and 16 models trained on equal-sized, non-overlapping segments of model-generated

data across 10 iterations. Ecosystems with lower diversity ($D; M = 1, 2$) perform well initially but exhibit rapidly increasing perplexity as the number of iterations grows. In contrast, the most diverse ecosystem ($M = 16$) starts with relatively high perplexity but improves over time. Across the full 10 iterations, a diversity level of four ($D; M = 4$) yields the lowest aggregated mean perplexity for both model families, with a stronger effect for OPT-125M. The least diverse ecosystem ($D = 1$) performs best only at $t = 0$, when ecosystems differ solely in sample size per model ($n = N/M$), as all models are trained on the original data[5].

Figure 10 in the Appendix extends the analysis to 20 iterations for OPT-125M, showing that the optimal diversity level continues to increase beyond 10 iterations.

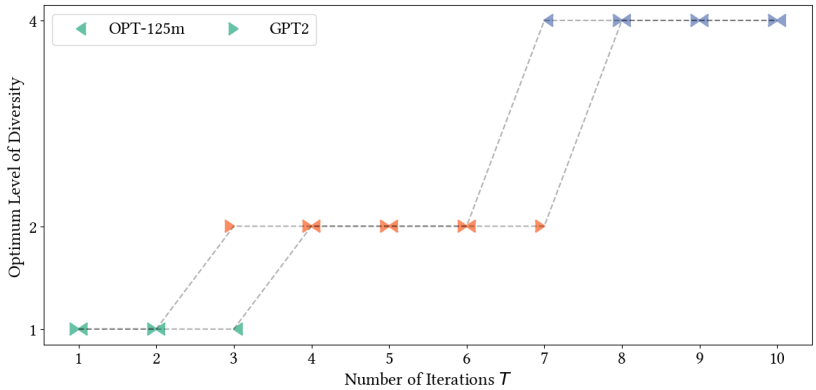

Fig. 3. **The optimal level of ecosystem diversity (Hill–Shannon diversity D) increases monotonically with the number of self-training iterations T.** The most homogeneous setting ($D = 1$, green) performs best only in the short term ($T \leq 3$ for OPT-125M and f$T \leq 2$ for GPT2). Thus, diversity D mitigates model collapse. As T grows, the data must be segmented across models to reach the diversity level that maximizes overall performance on the original test set ($D_{opt}(T) = \text{argmin}_D\, \mu_T(D)$). The corresponding performance trajectories are presented in Figure 4. Colored symbols correspond to the diversity levels shown in Figure 2 1, 2, 4, and 16 (e.g., orange denotes $D = 2$). Intermediate levels are omitted, as the key result is the monotonic increase in diversity benefits with T. An extended range of up to 20 iterations is shown in Figure 10 in the Appendix.

4.1 V1: Model size and data size

The benefits of diversity increase as both model size (from 125M to 350M parameters) and dataset size (from 0.8M to 2.1M tokens) scale (Figure 5). Holding dataset size constant, larger models exhibit a steeper increase in the optimal diversity level as the number of iterations grows. A similar pattern appears when increasing dataset size for the larger model, whereas for the smaller model dataset size has only a minor effect on the optimal diversity level. Below, we highlight several notable performance effects relative to the primary setting (125M-parameter model and 0.8M tokens dataset).

At iteration step $t = 0$, when models are trained on the original dataset, increasing either model or dataset size improves performance in line with established scaling laws [33, 37]. The most pronounced effects of larger models is the amplified collapse of low-diversity ecosystems ($D \leq 2$), leading to the observed shift in the optimal diversity level. While the effects of dataset size are

[5]The perplexity progression, including the perplexity of the original models, is shown in Figure 8 in the appendix.